Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	6.1818	5	5.1927	count
ring_length_r30_px	headline	399.273	403	153.861	px
ring_length_r60_px	reported	1055.18	1100	357.62	px
owned_length_px	ownership-dependent	2242.85	2419.61	1366	px
edge_count	ownership-dependent	101.364	112	63.0782	count
mean_edge_length_px	ownership-dependent	22.5943	21.6037	4.124	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	11901.2	12118	3231.07	px^2
ring_area_r60_px2	normalised	28359.4	29514	6840.82	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0327346	0.0321236	0.00693422	px^-1
ring_density_r60	normalised	0.0365721	0.037156	0.00790943	px^-1
roots_per_100px_perimeter	normalised	1.6497	1.6009	0.8214	per 100 px
ring_attached_length_r30_px	network v2	314.909	301	163.204	px
ring_attached_length_r60_px	network v2	786	794	380.794	px
ring_length_w_r30_px	network v2	444.7	443.978	175.736	px
ring_length_w_r60_px	network v2	1189.75	1240.18	409.865	px
sholl_crossings_r10	network v2	9.5455	9	5.2605	count
sholl_crossings_r20	network v2	14.1818	14	6.29	count
sholl_crossings_r30	network v2	15.8182	17	5.3818	count
ring_width_mean_r30_px	width	7.355	7.398	0.5346	px

Per field
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measure	value	units
analysed_area_px2	1.00717e+06	px^2
total_skeleton_length_px	36672	px
canalicular_length_density_per_px	0.036411	px^-1
junction_count	777	count
junction_density_per_px2	0.00077147	px^-2
lacuna_count	13	count
lacunae_per_px2	1.291e-05	px^-2
skeleton_component_count	247	count

mean_component_length_px	148.47	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0411402	px^-1
field_density_without_flagged_per_px	0.0363424	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	372	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.036411	px^-1
field_length_density_with_unattached_per_px	0.0367804	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0998	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	35180	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False
13								1015.14	933.902	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	219	584	872.482	48	18.1767	7.3357
2	20	684	1477	3155.21	160	19.72	7.4739
3	3	403	1085	739.659	23	32.1591	7.5955
4	10	488	1259	3854.17	173	22.2784	7.3773
5	6	424	1100	2973.74	152	19.5641	8.1648
6	5	385	1117	3650.71	159	22.9604	7.1407
7	5	535	1476	1773.79	84	21.1166	7.9782
8	4	338	972	2419.61	112	21.6037	7.3532
9	7	365	878	1474.51	69	21.3698	6.5411
10	5	448	1323	3831.3	152	25.2059	7.5279
11	1	223	629	16.4853	1	16.4853	7.398

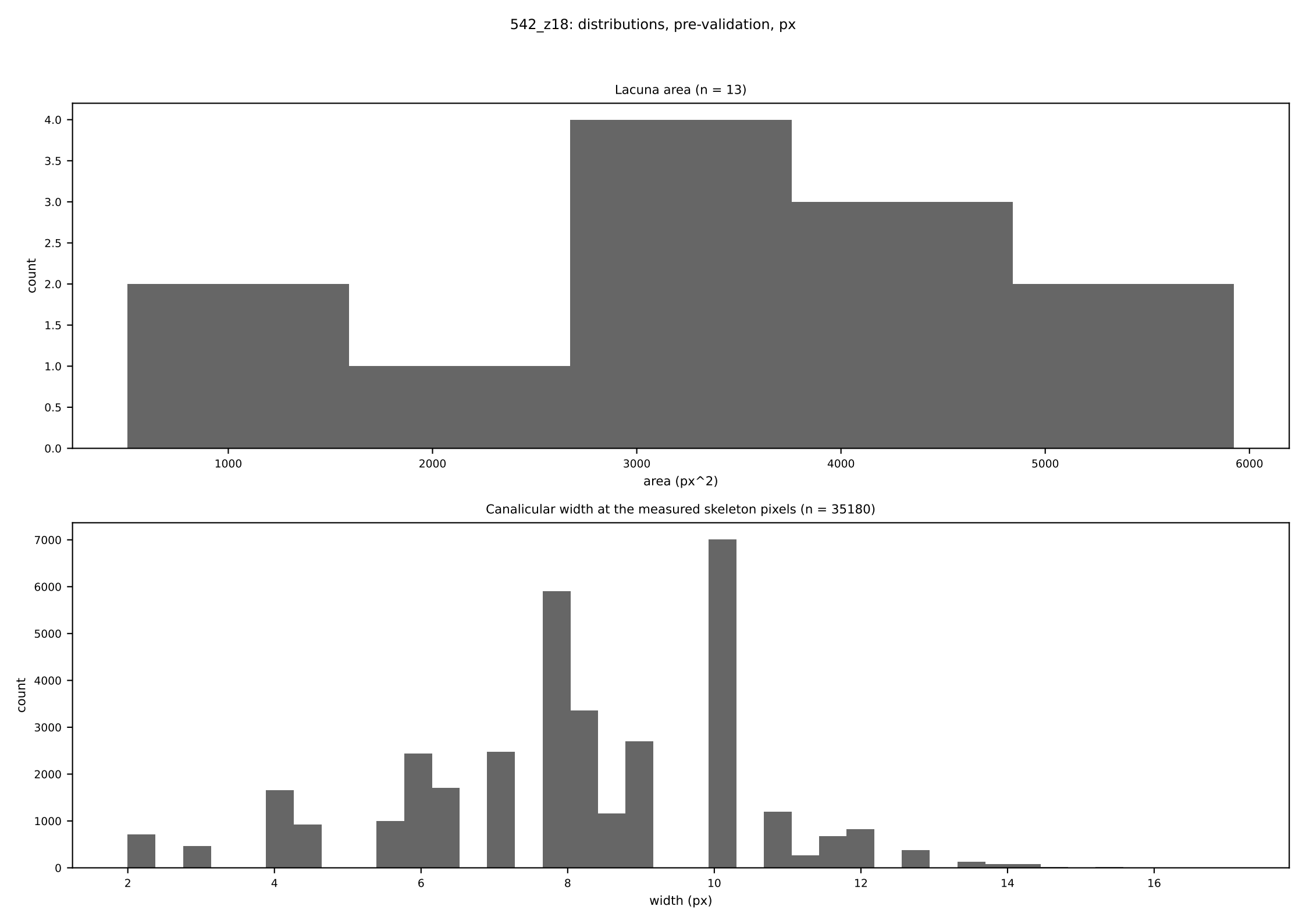
12	2	99	291	782.21	30	26.0737	6.3545	
13	0	0	9	0	0	0		

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0478688	0.0501675	1.4005
2	625.725	16860	35430	0.9542	0.8861	0.0405694	0.0416878	3.1963
3	377.019	13152	31926	1	1	0.0306417	0.0339848	0.7957
4	337.706	12118	29514	1	1	0.0402707	0.0426577	2.9612
5	371.806	13199	31872	1	0.9956	0.0321236	0.0345131	1.6137
6	308.593	11379	28385	1	1	0.0338343	0.0393518	1.6203
7	382.877	12760	30245	1	1	0.0419279	0.0488015	1.3059
8	306.492	10697	26160	1	1	0.0315976	0.037156	1.3051
9	437.262	14880	35351	1	1	0.0245296	0.0248366	1.6009
10	326.836	11837	29271	1	1	0.0378474	0.0451983	1.5298
11	263.48	9968	22850	0.9798	0.8783	0.0223716	0.0275273	0.3795
12	108.77	4063	10949	0.8837	0.75	0.0243662	0.0265778	1.8387
13	59.8995	971	1939	0.6158	0.5712	0	0.00464157	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	212	456	249.823	658.144	4	7	6
2	664	1234	774.097	1686.39	22	24	23
3	238	756	443.978	1207.65	8	14	18
4	365	853	541.404	1440.35	10	17	19
5	301	863	470.12	1240.18	5	14	17
6	294	794	429.363	1277.96	8	14	16
7	471	1307	601.416	1660.87	12	24	20
8	216	555	367.865	1080.25	9	13	15
9	329	651	415.463	998.909	12	9	13
10	390	1236	498.576	1479.74	12	16	20
11	124	231	245.723	700.328	5	8	8
12	72	166	103.698	314.652	2	3	5
13	0	0	0	10.0711	0	0	0


542_z18: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35180)
count
width (px)